

LKS SMK

Tingkat Provinsi Ke-XXII

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SOAL

BIDANG LOMBA

IT-Software Application


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**KEMENTERIAN PENDIDIKAN DAN KEBUDAYAN
DIREKTORAT JENDERAL PENDIDIKAN MENENGAH
DIREKTORAT PEMBINAAN SEKOLAH MENENGAH KEJURUAN**

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Module 1a:

Narrative / Description of Problem

You are required to make a system to provide information about the Convention on the *Kaltim* Party and Participant Selection Schedule.

Problem Criteria

As a party that has a high electability in Indonesia, the Kaltim Party wants to find the right Presidential candidate to represent the party in the presidential election of 2014. Kaltim Party requires a reliable management information system.

This system must be able to collect all the data concerning the Convention participants to be processed into useful information system for 'Capres' Determination Convention. You are required to build such systems with these requirements:

1. As the first task, you should build :
 - a. A Use Case Diagram
Design a use case diagram with the appropriate actors. The system should be designed to be able to provide detailed information that is required from each participant convention, presenting the number of votes (voting from party members and the society/people) and presents a selection schedule. Selection in this convention include an introduction to the public, media interviews, and debates among candidates.
 - b. An activity Diagram
Design an activity diagram with the corresponding activities. The main activities in this system are to present the selection timetable, the vote and convention participants and manage the participants data. Make sure all actors illustrated in the diagram and connected properly with the activity.
 - c. Entity Relationship Diagram (ERD)
Construct an appropriate ERD. Complete each entity with the appropriate attributes. Do not forget the key attributes and relation between entities.
Participant data should include: number of participant, name, place and date of birth, cadre/noncadre. Achievement data includes: name, year level, the hosting organization. Data includes: positions ever occupied the position, place, date of beginning, and the end date. While the selection schedule includes the date and place of execution.

Module 1b:

Candidate Number	First Name	Last Name	Current Position	Current Status	Place, Date of Birth
1	Abie	Murzaki	Head of DPR	Cadre	Palembang, November 6th, 1955
2	Irfan	Gustam	Head of DPD	noncadre	Padang Panjang, February 11th, 1962
3	Joni	Djalal	Indonesian Ambassador for USA	Cadre	Yugoslavia, September 10th, 1965
4	Ali	Musa	Member of BPK	noncadre	Tulungagung, September 12th, 1962
5	Dahlan	Isfan	BUMN Minister	noncadre	Magetan, August 17th, 1951
6	Rita	Wiriawan	Trade Minister	Cadre	Jakarta, September 21st, 1965
7	Ani	Baswid	Rector of Paramadina University	noncadre	Kuningan, May 2nd, 1969

Create a similar table on your Excel. For your information, this table contains raw data. You still have to break down these data before inserting in new tables that you create.

Create a database that contains the tables to store the data required.

This system requires data of convention participants such as: Participant Number, Name, place and date of birth, origin (cadre/non-cadre).

Achievement data include: Achievement Name, year, level, the hosting organization.

Position data were ever occupied include: position, place, date of beginning, end date. In addition, the data on the system is also able to keep track selection. Selection Convention include: an introduction to the public, media interviews and debates among candidates.

Schedule per candidate selection is stored in the system include: the date and place of execution.

You are asked to specify the most appropriate data type for each column needed. Make a complete database with primary key and foreign key corresponding, **you can add other fields if necessary.**

Then fill the data in the tables that you created by using data XLSX file that you have been created in the beginning (the raw data table). XLSX file data is divided into two:

Data A (containing all the data participants in conventions, achievements and positions ever occupied) and
Data B (containing all participant selection convention schedule).

Noteworthy is that both the data file contains raw data that must be processed (broken down) again before being inserted into the tables that you have created.

When all the tables have been filled by appropriate data, then make a user-friendly Display System in terms of the presentation of information, inserting, adding, deleting, and updating data.

Additional Terms:

- a) Each Primary Key should use the input mask
- b) The date format is: dd / mm / yyyy
- c) Each presentation of the data should has a searching facility.
- d) Debate Schedule between candidates displayed as a scheme.

Module 2:

Now, you should design a user interface that suitable for the UML diagrams, system function and database that you created in the previous modules.

The design of the interface can use the Visio Drawing (minimum) or Visual Studio (if possible).

The requirements to be met by the user interface are:

- a. consistent
 - b. Enable frequent users to use shortcuts
 - c. Offer informative feedback
 - d. Design dialog to yield closure
 - e. Offer simple error handling
 - f. Permit easy reversal of actions
 - g. Support internal locus of control
 - h. Reduce short-term memory load
- use case diagram translated into application form
 - all information is provided in the form field
 - clean and neat layout
 - use appropriate component

Module 3:

Narrative / Description of Problem

Kaltim party convention was held partially open and the selection will be carried out in a transparent manner. This Convention does not only involve the leader and members of the Kaltim party but also of the people in determining the winner of the election and conventions (*Capres*). Unfortunately, Kaltim party has difficulty to record the voting result from internal (*DPP, DPD and DPC*) and from public (society).

Problem Criteria

Please add another table to store the data members of the party include: number of member, name, section (*DPP, DPD or DPC*). Voting members data of the party include: number of member, which participant that he/she chooses, time, reason for choosing. Voting public data include: ID card number, name of the community, which participant that she/he chooses, time, and the reason for choosing.

1. Voting from Members of Party

Make a voting facility where party members can choose one of the participants of the convention. The voter should input his/her member number first. The voting process fails if the number entered is not registered or the number has already voted. Each member can only vote once. Election time is saved automatically.

2. Voting from public

Create a voting facility where the public can pick one candidate by entering his/her name first. Voting process fails if the ID card number has already voted. Each person can only vote once. Election time is saved automatically.

3. Documenting Voting Results

Create a facility to display the voting results of the data for each participant as in the table below:

Candidate ID	First name	Last Name	Votes from Party	Votes from Public	Total
K-01	Abie	Murzaki	12	890	902
BK-01	Irfan	Gustam	20	1010	1030
K-02	Joni	Djalal	38	2001	2049